COMP 2233 Data Structures — Laboratory Reference

Attachment A: ANSI/ISO Flowchart Symbols Guide

Flowcharts are standardized graphical representations of algorithms that use geometric shapes to depict computational steps, control flow, decision branches, and input/output operations. In accordance with **ISO/IEC 5807:1985**, this reference guide defines the core symbols required for Lab 2 (Experiment 1 & Experiment 2) and maps them directly to C programming constructs.

1. Core ANSI/ISO Flowchart Symbols Reference Table

Symbol Name	Shape / Geometry	Algorithmic Function	C Language Code Mapping
Terminal	Oval / Stadium	Marks the official START or STOP boundary of an algorithm or function.	int main() / return 0;
Process	Rectangle	Represents computational steps, variable assignments, or mathematical calculations.	int pos = 0; sum = a + b;
Decision	Diamond	Evaluates a logical condition. Has one entry flowline and two exit flowlines (True/False).	if (strcmp(a, b) == 0) while (i < N)
Input / Output	Parallelogram	Represents data entering or exiting the system (terminal I/O, user entry).	scanf("%15s", target); printf("%s", name);
Preparation / Loop	Hexagon	Defines variable initialization or loop counter control setup.	for (int i=0; i<N; i++)
Flowline	Arrow / Line	Indicates the exact execution path and direction of control logic.	Sequential statement flow
Connector	Small Circle	Connects complex flowlines across sub-diagrams without overlapping lines.	Function call / Jump point

2. Classification of Flowchart Structures

Algorithms are categorized into three primary structural archetypes based on their flowchart logic:

• **Linear Flowcharts:** Sequential execution from top to bottom with zero decision branching or loop feedback jumps (e.g., calculating simple formulas).

• **Logical Flowcharts:** Incorporate Decision Diamonds (if-else branching) to split execution paths based on runtime conditions (e.g., discrete variable checking).

• **Circular Flowcharts:** Contain loop feedback lines that re-evaluate conditions repeatedly until a termination condition is met (e.g., array iteration loops).

3. Guidelines for Lab 2 Flowchart Construction

Lab 2 Design Imperatives

Experiment 1 (Shopping List Search):

• *Discrete Approach:* Construct a cascading logical flowchart with 10 sequential Decision Diamonds checking item0 through item9.

• *Array Approach:* Construct a circular flowchart featuring an Initialization Hexagon (i = 0), a Decision Diamond (i < N), a Process Box for index increment (i++), and an internal Decision Diamond checking strcmp(list[i], target) == 0.

Experiment 2 (Passenger Manifest Search):

• Ensure struct attributes (pnr, name, seatNumber) are grouped in Output Parallelograms upon successful match.

• For Binary Search flowcharts, include Decision Diamonds splitting into three paths: cmp == 0 (Match), cmp < 0 (Search Left: high = mid - 1), and cmp > 0 (Search Right: low = mid + 1).